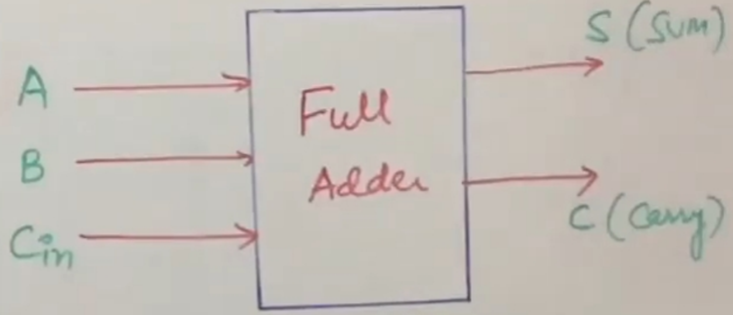
Designing of a Full Adder

$2^n = 2^3 = 8$

A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



=> Adds 3-inputs and generates 2-outputs

=> 1st two inputs are Standard I/Ps (A) & (B) and the third input is Carry in that are fed into the full Adder.

=> The two O/Ps are Sum & Carry.

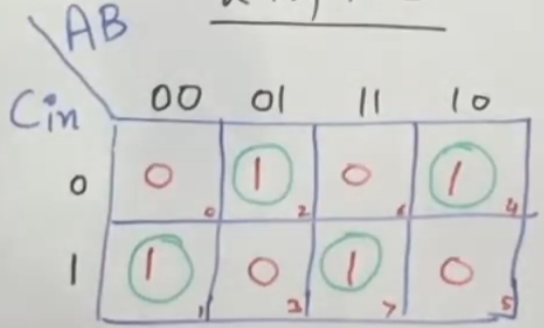
=> A Full Adder is indeed a 3-bit adder.

Designing of a Full Adder

$2^n = 2^3 = 8$

A	B	C_{in}	S	C
0	[illegible]	[illegible]	0	0
0	[illegible]	[illegible]	1	0
0	[illegible]	[illegible]	[illegible]	0
0	[illegible]	[illegible]	0	1
[illegible]	[illegible]	[illegible]	1	0
[illegible]	[illegible]	[illegible]	0	1
[illegible]	[illegible]	[illegible]	0	1
[illegible]	[illegible]	[illegible]	1	1

K-Map for S



C_{in} \ AB	00	01	11	10
0	0 (0)	1 (2)	0 (6)	1 (4)
1	1 (1)	0 (3)	1 (7)	0 (5)

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + AB\,C_{in} + A\bar{B}\bar{C}_{in}$$

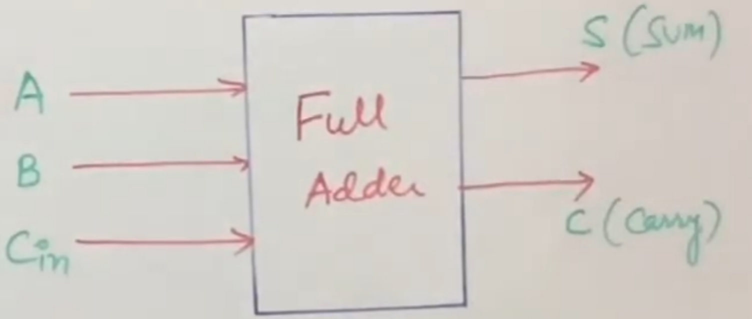
$$= \bar{A}\bar{B}C_{in} + AB\,C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in}$$

$$= (\bar{A}\bar{B} + AB)\,C_{in} + (\bar{A}B + A\bar{B})\,\bar{C}_{in}$$

$$= (\overline{A\oplus B})\,C_{in} + (A\oplus B)\,\bar{C}_{in}$$

$$= \bar{X}\cdot C_{in} + X\,\bar{C}_{in} \quad \{X = A\oplus B\}$$

$$S = X \oplus C_{in} = \boxed{A\oplus B\oplus C_{in}}$$



K-Map for C

C_{in} \ AB	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$C = AB + AC_{in} + BC_{in}$$

$$\boxed{C = AB + C_{in}(A \oplus B)}$$

$$c = AB + \bar{A}BC_{in} + A\bar{B}C_{in}$$

$$= AB + C_{in}(\bar{A}B + A\bar{B})$$

$$= AB + C_{in}(A \oplus B)$$

$+ A\bar{B}\bar{C}_{in}$

$- A\bar{B}\bar{C}_{in}$

$\bar{B})\bar{C}_{in}$

$A \oplus B]$

Designing of a Full Adder

$2^n = 2^3 = 8$

A	B	Cin	S	C
			0	0
			1	0
			1	0
			0	1
			1	0
			0	1
			0	
				1

K-Map for S

Cin \ AB	00	01	11	10
0	0 (0)	1 (2)	0 (6)	1 (4)
1	1 (1)	0 (3)	1 (7)	0 (5)

$S = A \oplus B \oplus Cin$

K-Map for C

Cin \ AB	00	01	11	10
0	0 (0)	0 (2)	1 (6)	0 (4)
1	0 (1)	1 (3)	1 (7)	1 (5)

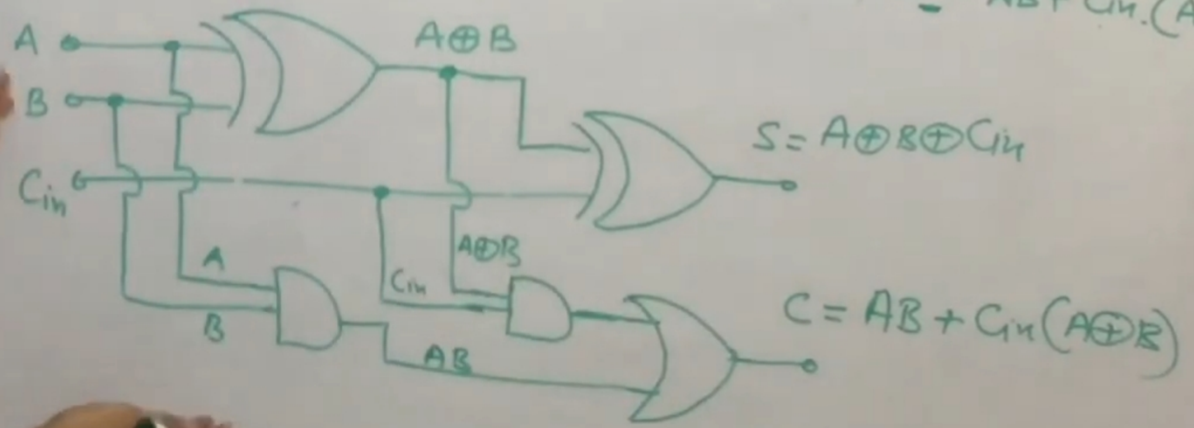
$C = AB + ACin + BCin$

$\boxed{C = AB + Cin(A \oplus B)}$

$C = AB + \bar{A}BCin + A\bar{B}Cin$

$= AB + Cin(\bar{A}B + A\bar{B})$

$C = AB + Cin.(A \oplus B)$



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